

2016 APMCM Problem C

Evaluation and Customization of Film and Television

For the past 50 years since the founding of the People's Republic of China, especially after the political reformation and economic opening up, China's radio and television media has achieved remarkable results and developed quickly along with the country's economic growth, scientific and technological developments, as well as living standard improvements. As of now, China's radio and television media has built a good foundation for future industrial development.

China produced a total of 893 films in 2012 with a total film box office of 17.073 billion CNY, which is a 30.18% increase from the previous year. Out of this amount, Chinese domestic film box office totaled to 8.273 billion CNY, which accounted for 48.46% of the annual total amount. This is the first time since 2003 that Chinese domestic films defeated by imported films. In February 2012, China and the United States reached an agreement on the WTO Memorandum of Understanding on issues related to the WTO films, and decided to increase the imported film quota. On top of the original 20-imported film quota, there will be 14 more high-tech formatted American films. The proportion of the box office sub-account also increased from the previous 17.5% to 25%. Domestic film market is in a bad position now.

The cultural industry in the United States which is actually known as the copyright industry, accounts for 10% of GDP, while it is only less than 3% in China. Nowadays in China, the price of a single episode of a TV show can go up to 7 digits, and it can probably reach 8 digits in the future. This is also in line with China's economic development. The increased demand for culture and entertainment after the affluent economy will drive and push the industry to grow rapidly and become one of the most fast-growing businesses in the future.

During the "12th Five-Year Plan" period, China will vigorously develop seven key cultural industries, such as film and television production, cultural-related creation, entertainment and performance. During this time, China's film and television industry will meet an important period of strategic opportunities; government supporting policy will be intensified in order to create a good business environment for Chinese film and television industry. Chinese film award committee will maintain an annual production of 500 films in the next 5 years. Chinese movie market will exceed 30 billion CNY at the box office in 2014 and the number of film viewers is expected to reach 1.112 billion, which is quite close to the total national population. In addition, cinemas will become more diversified; niche and art films will have their own screening space as well. The State Council's "Film Industry Prosperity and Development Promotion Guidance," pointed out that by the end of 2015, through reformation and innovation, increased investment and accelerated development, China will establish and improve fair market competition, enterprise and independent operation of the film industry operating system, market operation, business management, government procurement, and the public benefit of the film public service system.

The current Chinese television market is highly competitive and full of different styles and topics.

Though the annual TV drama production remains high, but TV stations do not have much desire to buy these dramas, which leads to the oversupply of TV dramas. According to statistics, more than half of all TV productions each year go to waste without a buyer post-production, creating in a massive waste of resources. Quantity has never been a problem for TV dramas, yet quality has always been the problem we need to solve. How to lower cost and get rid of the hasty and crude TV investment hold the future development for TV dramas. The future competition between TV dramas will be a race of quality. It is only possible to receive maximum return by ensuring the quality of TV dramas.

In the meantime, using Big Data from 2014 as an analysis instrument to test TV market has been quite successful. Though Big Data cannot create the script, it can analyze data and forecast pretty precisely. This could be applied to script writing, TV rating forecast, outcome of TV commercials, and TV drama purchase. It is possible to reduce TV investment risk, improve script quality, and forecast audience response to ensure maximum benefits.

In film and television drama market, how to evaluate and customize the film and television drama and other issues has always been the center of focus during a production. Now please try to use mathematical modeling methods to solve the following problems.

Question 1: Rank the TV dramas based on the ranking index and name your top 10 with the data given in Annex 1 and Annex 2.

Question 2: Actor/actress's popularity ranking could be very useful to a TV series production. Please collect and use related data as foundation, design a star popularity index, and try to prove the attainability of your index by giving a real-life example from this year.

Question 3: Based on common opinion data (such as the number of micro blogging reviews, Baidu search volume, Youku clicks, etc.) and the current audience to speculate on viewers' interest and what would make a TV show popular. Describe an ideal production team, including the producer and actors. Try to prove your point with a real-life example.

Question 4: How to find and recommend the most interesting and suitable TV drama for each viewer based on his/her viewing history? How to find the script content most suitable for each local TV station according to the ratings of programs broadcasted by the local TV stations is also worth looking into. Collect related data, provide a solution by using mathematical modeling methods, and use a real-life example to prove your point.